5. Evaluate the Wronskian $W(y_1, y_2, y_3)(0)$, where $y_1(x) = 1, y_2(x) = e^x, y_3(x) = e^{-x}$

(a) -2
(b) -1
(c) 0
(d) 2
(e) None of the above

6. Let $y(t)$ be the solution of the initial value problem

$$y'' + 2y' - 3y = 0 \quad y(0) = b \quad y'(0) = 3$$

For which value of b is $\lim_{t\to\infty} y(t) = 0$?

(a) No value for b with the limit 0
(b) 0
(c) -1
(d) 1
(e) None of the above

7. Which of the following second order homogeneous linear equations has $y_1(t) = e^{2t}$ and $y_2(t) = e^{-3t}$ as two of its solutions?

(a) $2y'' + 2y' - 12y = 0$
(b) $y'' + 5y' + 6y = 0$
(c) $y'' - y' - 6y = 0$
(d) $2y'' - 3y' = 0$
(e) None of the above

8. Find the differential equation whose general solution is given by $y = \sin(Ax + B)$

(a) $y^2y'' - (y')^2 = 0$
(b) $(1 - y^2)y'' - y(y')^2 = 0$
(c) $(y'')^2 - y^2y' = 0$
(d) $y^2y'' - (y'/y)^2 = 0$
(e) None of the above

9. Find the general solution of the differential equation: $y''' + y' = 1$

(a) $y = A + B\cos t + C\sin t + t/2$
(b) $y = A + B\cos t + C\sin t + 1$
(c) $y = A + B\cos t + C\sin t + t$
(d) $y = A + B\cos t + C\sin t + 2$
(e) None of the above

10. Find the solution of the initial value $X'(t) = AX(t)$ with

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \text{ and } X(0) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

(a) $X(t) = \begin{bmatrix} -e^t + e^{3t} \\ e^t + e^{3t} \end{bmatrix}$
(b) $X(t) = \begin{bmatrix} e^{3t} \\ e^{3t} \end{bmatrix}$
(c) $X(t) = \begin{bmatrix} -e^t \\ e^t \end{bmatrix}$
(d) $X(t) = \begin{bmatrix} -e^t - e^{3t} \\ e^t - e^{3t} \end{bmatrix}$
(e) None of the above

11. Solve the differential equation

$$y' + \tan(x)y = \cos^2(x), \quad y(0) = 2$$

(a) $y = (\sin(x) - 2)\cos(x)$
(b) $y = (\sin(x) + 2)\cos(x)$
(c) $y = (\cos(x) + 2)\sin(x)$
(d) $y = (\cos(x) - 2)\sin(x)$
(e) None of the above

12. Solve the differential equation

$$\frac{dy}{dx} = \frac{y^2 - 1}{x}; \quad y(1) = 2$$

(a) $y = \frac{3-x^2}{3+x^2}$
(b) $y = \frac{1+3x^2}{1-x^2}$
(c) $y = \frac{1-3x^2}{1+x^2}$
(d) $y = \frac{3+x^2}{3-x^2}$
(e) None of the above

Consider the differential equation below and use it to solve question 13 - 17

$$\begin{aligned} x_1' &= x_1 + 2x_2 \\ x_2' &= 2x_1 + x_2 \end{aligned}$$

13. The eigenvalues of the corresponding to the system are
    (a) 1, -1
    (b) 2, -1
    (c) 3, -1
    (d) 4, 2
    (e) None of the above

14. The eigenvectors associated with theses eigenvalues are
    (a) $(1,1)^T, (2,2)^T$
    (b) $(2,-1)^T, (1,-1)^T$
    (c) $(2,1)^T, (1,-1)^T$
    (d) $(1,-1)^T, (1,1)^T$
    (e) None of the above

15. The general solution to the system of differential equation is given by
    (a) $X = Ae^t(1,1)^T + Be^{-t}(2,2)^T$
    (b) $X = Ae^{2t}(2,-1)^T + Be^{-t}(1,-1)^T$
    (c) $X = Ae^{3t}(2,1)^T + Be^{-t}(1,-1)^T$
    (d) $X = Ae^{4t}(1,-1)^T + Be^{2t}(1,1)^T$
    (e) None of the above

16. The fundamental matrix of the system of differential equations is given by
    (a) $\phi(t) = \begin{bmatrix} e^t & 2e^{-t} \\ e^t & 2e^{-t} \end{bmatrix}$
    (b) $\phi(t) = \begin{bmatrix} 2e^{3t} & e^{-t} \\ e^{3t} & -e^{-t} \end{bmatrix}$
    (c) $\phi(t) = \begin{bmatrix} 2e^{3t} & e^{-t} \\ -e^{3t} & -e^{-t} \end{bmatrix}$
    (d) $\phi(t) = \begin{bmatrix} e^{4t} & e^{2t} \\ -e^{3t} & e^{2t} \end{bmatrix}$
    (e) None of the above

17. Given the initial condition $X(0) = (3,0)^T$, constant A, B in the general solution of the system of differential equation takes the values?
    (a) 1, 1
    (b) 2, -1
    (c) 3, 3
    (d) 4, 5
    (e) None of the above

18. Given $N(x,y)dy = -M(x,y)dx$ is not exact and $g(x) = \left(-\frac{dN}{dx} + \frac{dM}{dy}\right)/N$ then the integrating factor is
    (a) $-e^{\int g(x)dx}$
    (b) $-\int g(x)dx$
    (c) $\int g(x)dx$
    (d) $e^{\int g(x)dx}$
    (e) None of the above

19. Find the differential equation whose solution is $ye^x = x + c$
    (a) $y' + y = e^{-x}$
    (b) $y' = e^{-x}$
    (c) $y' + y = e^x$
    (d) $y'' + y' = e^{-x}$
    (e) None of the above

KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY

COLLEGE OF ENGINEERING

END OF SEMESTER EXAMINATIONS

(FIRST SEMESTER 2013)

[redacted]

INDEX NUMBER: _____

Instructions:

ANSWER ALL QUESTIONS.

PROVIDE ALL ANSWERS ON THE SCANNABLE SHEET PROVIDED.

MAKE SURE YOUR INDEX NUMBER IS WRITTEN CORRECTLY ABOVE AND ON THE SCANABLE SHEET AS WELL.

SUBMIT THE QUESTION PAPER AND THE SCANABLE SHEET AT THE END OF THE EXAMINATION

All the best!

1. Which substitution will reduce the nonlinear differential equation

$$x^2y' + 2xy = y^3$$

to linear differential equation

(a) $v = y^{-1}$

(b) $v = y^{-2}$

(c) $v = y^{-3}$

(d) $v = y^{-4}$

(e) None of the above

2. Find the value of b for which the differential equation is exact

$$(xy^2 + bx^2y)dx + (x + y)x^2dy = 0$$

(a) 1

(b) 2

(c) 3

(d) 4

(e) None of the above

3. Solve the differential equation

$$y' - 2y = x^2e^{2x}$$

(a) $y = \left(A + \frac{x^3}{3}\right)e^{-2x}$

(b) $y = \left(A - \frac{x^3}{3}\right)e^{2x}$

(c) $y = \left(A - \frac{x^3}{3}\right)e^{-2x}$

(d) $y = \left(A + \frac{x^3}{3}\right)e^{2x}$

(e) None of the above

4. Solve the differential equation

$$\frac{dy}{dx} = -2 - y + y^2$$

where $y_1 = 2$ is a particular solution.

(a) $y = \left(\frac{1}{3} + ce^{-3x}\right)^{-1}$

(b) $y = 2 + \left(\frac{1}{3} + ce^{-3x}\right)$

(c) $y = 2 - \left(\frac{1}{3} + ce^{-3x}\right)^{-1}$

(d) $y = -\left(\frac{1}{3} + ce^{-3x}\right)$

(e) None of the above

20. Determine the Wronskian of the solution $y = Ae^x + Be^{2x}$

(a) e^{3x}

(b) e^{-3x}

(c) $-e^{3x}$

(d) $-e^{-3x}$

(e) None of the above

21. Consider the differential equation $(2x + 3y - 3) + (3x + 2y - 7)\frac{dy}{dx} = 0$, determine the form of the substitution for x and y

(a) $x = X + 3,\ y = Y + 1$

(b) $x = X - 3,\ y = Y + 1$

(c) $x = X - 3,\ y = Y - 1$

(d) $x = X + 3,\ y = Y - 1$

(e) None of the above

22. Solve the differential equation $(4x + 8y - 12) + (8x + 16y + 20)\frac{dy}{dx} = 0$

(a) $(4x + 8y)^2 - 96x + 160y = A$

(b) $(4x + 8y) - 24x + 40y = A$

(c) $(x + 8y)^2 - 96x + 160y = A$

(d) $(x + 8y)^2 - 24x + 40y = A$

(e) None of the above

23. Which of the following differential equations is a Bernoulli's equation?

(a) $xy' + 2y = xe^x$

(b) $(x - y + 2)y' = x$

(c) $x^2y' + 2y = y^3$

(d) $yx^2y' + 2y = y^2$

(e) None of the above

24. Using the method of variation o f parameters, the parameters of the differential equation $y'' + y = \sec x$ are given by

(a) $u_1' = 0,\ u_2' = \sec x$

(b) $u_1' = \cos x,\ u_2' = \sin x$

(c) $u_1' = \tan x,\ u_2' = \sec x$

(d) $u_1' = \tan x,\ u_2' = 1$

(e) None of the above

25. Using the method of variation of parameters, find the general solution of the differential equation $y'' + y = \sec x$

(a) $(c_1 + \ln|\cos x|)\cos x + (c_2 + x)\sin x$

(b) $(c_1 \ln|\cos x|)\cos x + (c_2 x)\sin x$

(c) $(c_1 \tan x)\cos x + (c_2 \ln x)\sin x$

(d) $(c_1 \sec x) + (c_2 \ln x)cosecx$

(e) None of the above

26. Using the method of variation of parameters or otherwise, find the general solution of the differential equation $y'' + 4y' + 4y = xe^{2x}$

(a) $c_1xe^{2x} + c_2e^{-2x}$

(b) $\left(\frac{2x-1}{32}\right) + c_1xe^{2x} + c_2e^{-2x}$

(c) $(c_1 + c_2x)e^{-2x}$

(d) $\left(\frac{2x-1}{32}\right)e^{2x} + (c_1 + c_2x)e^{-2x}$

(e) None of the above

27. The nonlinear first order differential equation $y' + y = (xy)^2$ can be made into linear differential equation of the form

(a) $u' - u = -x^2$

(b) $u' + u = -2x$

(c) $u' + u = 0$

(d) $u' - u = 1 - x^2$

(e) None of the above

28. The solution of the nonlinear first differential equation $y' + y = (xy)^2$ is thus

(a) $y = 1 - x_A^2$

(b) $y = (2 + 2x + x^2 + Ae^x)^{-1}$

(c) $y = 2 + 2x^{-1} + x^{-2}Ae^{-x}$

(d) $y = A(1 - x^2)e^{-2x} + 2x$

(e) None of the above

29. Solve

$$x' = \begin{bmatrix} 2 & 2 \\ 0 & 2 \end{bmatrix} x + \begin{bmatrix} 4 \\ e^{2t} \end{bmatrix}$$

(a) $\begin{bmatrix} -2 + t^2e^{2t} \\ e^{2t} \end{bmatrix}$

(b) $\begin{bmatrix} t^2e^{2t} \\ e^{2t} \end{bmatrix}$

(c) $\begin{bmatrix} 2 + t^2e^{2t} \\ e^{2t} \end{bmatrix}$

(d) $\begin{bmatrix} -2 + t^2e^{2t} \\ te^{2t} \end{bmatrix}$

(e) None of the above

30. Suppose that a river is $2a = 1$ miles wide and that its midstream velocity is $v_0 = 9mi/h$. If the swimmer's velocity is $v_s = 3mi/h$. Assume that the trajectory of the swimmer satisfies the differential equation

$$\frac{dy}{dx} = \frac{v_0}{v_s}\left(1 - \frac{x^2}{a^2}\right)$$

How far does the swimmer drift when he crosses the river?

(a) 0.5 mile
(b) 1 mile
(c) 1.5 miles
(d) 2 miles
(e) None of the above

31. One of the following equations exact. Identify which one and find a solution $y(x)$ of that equation satisfying $y(0) = 0$. What is $y(1)$?

$$(2x+2y)+(2x+2y)\frac{dy}{dx} = 0; \quad (2x+3y)+(2x+3y)\frac{dy}{dx} = 0$$

(a) -2
(b) -1
(c) 0
(d) 1
(e) None of the above

Classify each of the first-order differential equation in Questions 32 - 36 in one or more of the following categories:

I. Linear
II. Variable Separable
III. Homogeneous Coefficeint of differentials
IV. Exact
V. Equation of the form $yf(x,y)dx + xg(x,y)dy = 0$

32. $(xy^2 + y)dx + (x^2y - x)dy = 0$
(a) I, IV
(b) IV, v
(c) v
(d) I, II, IV, V
(e) None of the above

33. $xdx + \frac{1}{y}dy = 0$
(a) I, II
(b) I. II, IV
(c) I, II, III, IV
(d) II, III, IV
(e) None of the above

34. $y^2dx + x^2dy = 0$
(a) I, II, III, IV
(b) II, III, IV
(c) III,IV
(d) II, III
(e) None of the above

35. $\left[x\sin(\frac{y}{x}) - y\cos(\frac{y}{x})\right]dx + x\cos(\frac{y}{x})dy = 0$
(a) III
(b) I, III
(c) II, III
(d) IIIM IV
(e) None of the above

36. $y' = \frac{-xy^2}{x^2y+y^2}$
(a) IV
(b) IV, V
(c) III, IV, V
(d) II, III, IV, V
(e) None of the above

37. Determine the degree of the differential equation

$$\left[\frac{d^2y}{dx^2} + \frac{dy}{dx}\right]^{\frac{1}{5}} = \left[\frac{dy}{dx} + \sin^2(x)\right]$$

(a) 1
(b) 3
(c) 5
(d) 10
(e) None of the above

38. The following differential equations are linear except

(a) $y'' - 3y' + 2y = 0$
(b) $\frac{d^2y}{dx^2} + \frac{d^2y}{y^2} + \frac{d^2y}{dx^2}$
(c) $xdx + ydy$
(d) $L\frac{di}{dt} + Ri = E$
(e) None of the above

39. Find the integrating factor of the differential equation

$$\left(3x + \frac{6}{y}\right)dx + \left(\frac{x^2}{y} + 3\frac{x}{y^2}\right)dy = 0$$

(a) $\mu(x, y) = x^2y$
(b) $\mu(x, y) = xy^2$
(c) $\mu(x, y) = xy$
(d) $\mu(x, y) = x^2 + y$
(e) None of the above

40. The differential equation

$$\left(\frac{\sin y}{y} - 2e^{-x}\sin x\right)dx + \left(\frac{\cos y + 2e^{-x}\cos x}{y}\right)dy = 0$$

, is not exact, but becomes exact when multiplied by the given integrating factor, $\mu(x, y) = ye^x$. Then solve the differential equation.

(a) $e^x \sin y + y\cos x = c$
(b) $e^x \sin y + y^2\cos x = c$
(c) $e^x \cos y + 2y\cos x = c$
(d) $e^x \sin y + 2y\cos x = c$
(e) None of the above

41. Find the particular solution of the following differential equation $y'' - 4y' - 12y = te^{4t}$

(a) $y_p = \frac{7}{36}(3t - 1)e^{4t}$
(b) $y_p = \frac{1}{36}(3t + 1)e^{4t}$
(c) $y_p = -\frac{1}{36}(3t + 1)e^{4t}$
(d) $y_p = -\frac{2}{3}(3t - 1)e^{4t}$
(e) None of the above

42. If the integrating factor of the differential equation $3x^2y^2dx + (2x^3y + x^3y^4)dy = 0$ is $e^{\frac{y^3}{3}}$, that will make the equation exact, what is the general solution of the differential equation?

(a) $x^3 + y^2e^{\frac{y^3}{3}} = c$
(b) $x^3y^2e^{\frac{y^3}{3}} = c$
(c) $x^3 - y^2e^{\frac{y^3}{3}} = c$
(d) $xy^2 + y^2e^{\frac{y^3}{3}} = c$
(e) None of the above

43. The population of a certain country is known to increase at a rate proportional to the number of people presently living in the country. If after two years the population has doubled and after three years the population is 50,000, estimate the number of people initially living in the country.

(a) 18000
(b) 17800
(c) 17700
(d) 17655
(e) None of the above

44. Let $y = f(x)$ be the solution of the equation

$$\frac{dy}{dx} = \frac{x^2}{x^2 + 1}$$

such that $y = 0$ when $x = 0$. What is the value of $f(1)$?

(a) $1 - \log 2$
(b) $1 + \log 2$
(c) $\log 2$
(d) $\frac{1}{4}(4 - \pi)$
(e) None of the above

45. Let $y = f(x)$ be the solution of the equation

$$\frac{dy}{dx} + \frac{y}{x} = \sin x$$

such that $f(\pi) = 1$. What is the value of $f(\frac{1}{2}\pi)$?

(a) $\frac{2}{\pi} - 1$
(b) $\frac{2}{\pi}$
(c) $\frac{2}{\pi} + 1$
(d) $\frac{\pi}{2}$
(e) None of the above

46. Solve for y, $(D^2 - 2D + 1)y = 2xe^x$

(a) $y = c_1e^x + c_2xe^x + x^3e^x$
(b) $y = c_1e^x + c_2xe^x + \frac{x^2}{3}e^x$
(c) $y = c_1e^x + c_2xe^x + \frac{x^3}{4}e^x$
(d) $y = c_1e^x + c_2xe^x + \frac{x^3}{3}e^x$
(e) None of the above

47. Write done the trial function, y_p form of the particular solution to $y'' + 8y' + 16y = e^{-4t} + (t^2 + 5)e^{-4t}$

(a) $y_p = t^2(At^2 + Bt + C)e^{-4t}$, where A, B, C are the coefficients to be determined
(b) $y_p = t(At^2+Bt+C)e^{-4t}$, where A, B, C are the coefficients to be determined
(c) $y_p = (At^2+Bt+C)e^{-4t}$, where A, B, C are the coefficients to be determined
(d) $y_p = t^2(At^2 + Bt)e^{-4t}$, where A, B, C are the coefficients to be determined
(e) None of the above

48. What is the Wronskian of the system

$$x' = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} x,$$

whose linear solutions are $x_1 = [e^{2t}, e^{2t}, e^{2t}]^T$, $x_2 = [e^{-2t}, 0, e^{-2t}]^T$ and $x_3 = [0, e^{-t}, -e^{-t}]^T$

(a) 0
(b) e^{-t}
(c) $-e^{-t}$
(d) 3
(e) None of the above

49. Which of the following is a general solution of the system:

$$\begin{aligned} (D-4)x + 3y &= 0 \\ -6x + (D+7)y &= 0 \end{aligned}$$

(a) $x = c_1e^{2t}+3c_2e^{-5t}$, $y = c_1e^{2t}+2c_2e^{-5t}$
(b) $x = c_1e^{2t}-3c_2e^{-5t}$, $y = c_1e^{2t}+2c_2e^{-5t}$
(c) $x = c_1e^{2t}+3c_2e^{-5t}$, $y = c_1e^{2t}-2c_2e^{-5t}$
(d) $x = c_1e^{2t} + c_2e^{-5t}$, $y = c_1e^{2t} + c_2e^{-5t}$
(e) None of the above

50. Solve $y'' + 2y' + 5y = 0$

(a) $y = c_1e^x \cos 2x + c_2e^x \sin 2x$
(b) $y = c_1e^x \cos 2x - c_2e^x \sin 2x$
(c) $y = c_1 \cos 2x + c_2e^x \sin 2x$
(d) $y = c_1e^x \cos 2x + c_2 \sin 2x$
(e) None of the above

51. Which of the following about $(y^2-y)dx+xdy = 0$ is true?

(a) It is first order linear differential equation
(b) It is non-homogeneous
(c) It is exact
(d) $y = c_1x^2 + \cos x$
(e) None of the above

52. Solve $(y - x^2)dx - (y + x)dy = 0$

(a) $3y^2 + 6xy - 2x^3 = c$
(b) $y^2 + xy - x^2 = c$
(c) $3y^2 + 6xy - x^2 = c$
(d) $3y^2 - 6xy + 2x^3 = c$
(e) None of the above

53. Which of the following differential equations can be solved by separating the variables?

(a) $x\frac{dy}{dx} = y + x$
(b) $xy\frac{dy}{dx} = x + 1$
(c) $x + \frac{dy}{dx} = \ln y$
(d) $e^{x+y} = y\frac{dy}{dx}$
(e) None of the above

54. Which of the following expressions is homogeneous in x and y?

(a) $x^2 - 3xy + y$

(b) $x^2 - 3xy + y^2$

(c) $x^3y - 3xy^2 + y^2$

(d) $x^3y - xy$

(e) None of the above

55. Find the value of the definite integral $\int_{-\pi}^{\pi} \cos 2x \sin 2x dx$

(a) 0

(b) 1

(c) -1

(d) 2

(e) None of the above

56. What is the period of $y = A\cos 2x$?

(a) 2

(b) 1

(c) 2π

(d) π

(e) None of the above

57. Find the Fourier series representation of $f(x) = x$ on the interval $-2 \leq x \leq 2$.

(a) $-\frac{4\cos n\pi}{n\pi}$

(b) $\frac{4\cos n\pi}{n\pi}$

(c) $-\frac{4\sin n\pi}{n\pi}$

(d) $\frac{4\sin n\pi}{n\pi}$

(e) None of the above

Use the following Differential Equation below with variable co-efficient to answer Questions 58 - 60

58. Determine all possible singularity of the differential equation. $x(1-x^2)\frac{d^2y}{dx^2} - 2\frac{dy}{dx} - 3y = 0$.

(a) $\pm 1, 0$

(b) $-2, -3$

(c) $2, -3$

(d) $0, -3$

(e) None of the above

59. What is the lower bound for the radius of convergence of series solution about $x = 3$

(a) $|x - 3| < 2$

(b) $|x + 3| < 2$

(c) $|x - 3| > 4$

(d) $|x + 3| > 4$

(e) None of the above

60. The O.D.E has the series of the form

(a) $\sum_{n=0}^{\infty} a_n x^n$

(b) $\sum_{n=0}^{\infty} a_n rx^n$

(c) $\sum_{n=0}^{\infty} a_n x^{n+m}$

(d) $\sum_{n=0}^{\infty} a_n rx^{n+m}$

(e) None of the above

KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY

COLLEGE OF ENGINEERING

MID-SEMESTER EXAMINATIONS

(FIRST SEMESTER 2014)

TIME: 1 HRS

Math 251: DIFFERENTIAL EQUATIONS

BSc. (ENGINEERING II)

INDEX NUMBER: [redacted] PROGRAMME: GEOMATIC ENGINEERING

Instructions:

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All the best!

1. Solve the differential equation given below:

$$(x+1)y'' - (x+2)y' + y = 0; \quad y_1(x) = x^3$$

(a) $y = c_1x^3 + c_2x^4 + 5e^x$

(b) $y = c_1x^3 + c_2x^4 - 5e^x$

(c) $y = c_1x^3 - c_2x^4 - 5e^x$

(d) $y = c_1x^3 + c_2x^4 + e^x$

(e) None of the above

2. Solve the differential equation given that one of the solutions is $y_1(x) = \sin x$:

$$y' = \frac{2\cos^2 x - \sin^2 x + y^2}{2\cos x}, \quad y(0) = -1$$

(a) $y = \frac{2\sec x}{-2-\tan x} - 2\sin x$

(b) $y = \frac{2\sec x}{-2-\tan x} - \sin x$

(c) $y = \frac{2\sec x}{-2-\tan x} + 2\sin x$

(d) $y = \frac{2\sec x}{-2-\tan x} + \sin x$

(e) None of the above

3. Solve the differential equation given below:

$$(3x - y + 1)dx + (2y + 3 - 6x)dy = 0$$

(a) $\ln|5x - 10| + 2x = 5x + C$
(b) $\ln|2x - 10| + 2x = 5x + C$
(c) $\ln|5x - 10| + x = 5x + C$
(d) $\ln|2x - 10| + x = 5x + C$
(e) None of the above

4. The functions $f(x)$ and $g(x)$ as provided below as said to be:

$$f(x) = 9\cos 2x \quad g(x) = 2\cos^2 x - 2\sin x$$

(a) Linearly dependent
(b) Linear Differential Equation
(c) Linearly independent
(d) Initial Value Problem
(e) None of the above

5. Solve the initial value problem provided below:

$$\frac{dr}{dt} = -4rt, \text{ when } t = 0, \ r = r_0$$

(a) $r = r_0 e^{-2t}$
(b) $r = r_0 e^{2t^2}$
(c) $r = r_0 e^{-2t^2}$
(d) $r = re^{2t^2}$
(e) None of the above

6. Solve the given differential equation:

$$y' + y = \sin x$$

(a) $ce^{-x} + \frac{1}{2}\sin x, + \frac{1}{2}\cos x$
(b) $ce^{-x} - \frac{1}{2}\sin x - \frac{1}{2}\cos x$
(c) $ce^{-x} + \frac{1}{2}\sin x - \frac{1}{2}\cos x$
(d) $ce^{-x} + \frac{1}{2}\sin x + \frac{1}{2}\cos 2x$
(e) None of the above

7. Solve the differential equation given be

$$xy\frac{dy}{dx} = x^2 + y^2 \quad y(1) = -2$$

(a) $y^2 = 2x^2(\ln|x| - 2)$
(b) $y^2 = x^2(\ln|x| + 2)$
(c) $y^2 = 2x^2(\ln|x| + 2)$
(d) $y^2 = x^2(\ln|x|2)$
(e) None of the above

8. Which of the following represent the
erential for of the equation below?

$$\frac{dy}{dx} = \frac{y}{x}$$

(a) $ydx - xdy = 0$
(b) $ydx + xdy = 0$
(c) $ydx - xdy = 2$
(d) $ydx + xdy = 2$
(e) None of the above

9. Solve the differential equation:

$$9y\frac{dy}{dx} + 4x = 0$$

(a) $y^2 + \frac{4x^2}{9} = \frac{2c}{9}$

(b) $y^2 + \frac{4x^2}{9} = \frac{2c}{5}$

(c) $y^2 + \frac{4x^2}{5} = \frac{2c}{9}$

(d) $y + \frac{4x^2}{9} = \frac{2c}{9}$

(e) None of the above

10. Solve the differential equation given that one of the solutions is $y_1(x) = x$:

$$y' = -y^2 + xy + 1$$

(a) $y = \frac{x}{cxe^{\frac{1}{2}x^2} - 1} + 2x$

(b) $y = \frac{x}{cxe^{\frac{1}{2}x^2} - 1} + x$

(c) $y = \frac{2x}{cxe^{\frac{1}{2}x^2} - 1} + 2x$

(d) $y = \frac{x}{cxe^{\frac{1}{2}x^2} - 1} - x$

(e) None of the above

11. Solve the differential equation:

$$(D^2 - 9)y = e^{3x}$$

(a) $y = \frac{xe^{3x}}{6} - \frac{e^{3x}}{36} + \frac{c_1e^{3x}}{6} + c_2e^{3x}$

(b) $y = \frac{xe^{3x}}{6} - \frac{e^{3x}}{36} + \frac{c_1e^{3x}}{6} + c_2e^{-3x}$

(c) $y = \frac{xe^{3x}}{6} - \frac{e^{3x}}{36} + \frac{c_1e^{3x}}{6} + c_2e^{-3x}$

(d) $y = \frac{xe^{3x}}{6} - \frac{e^{3x}}{3} + \frac{c_1e^{3x}}{6} + c_2e^{-3x}$

(e) None of the above

12. Solve the given initial value equation:

$$y' + \frac{2y}{x} = -x^9y^5, \quad y(-1) = 2$$

(a) $y^{-4} = 2x^{10} - \frac{31}{16}x^8$

(b) $y^{-4} = 2x^{10} + \frac{31}{16}x^8$

(c) $y^{-4} = -2x^{10} - \frac{31}{16}x^8$

(d) $y^4 = 2x^{10} - \frac{31}{16}x^8$

(e) None of the above

13. Solve the differential equation given below:

$$(6x+4y+1)dx+(4x+2y+2)dy = 0, \quad y(1/2) =$$

(a) $12x^2 + 4x + 16xy + 8y + 4y^2 = -$

(b) $12x^2 + 4x + 16xy + 8y + 4y^2 =$

(c) $12x^2 + 4x + 16xy + 8y + y^2 = -$

(d) $2x^2 + 4x + 16xy + 8y + 4y^2 =$

(e) None of the above

14. Solve the differential equation given below:

$$xyy' = 2y^2 + x^2$$

(a) $y = x\sqrt{2x^2 \quad 2}$

(b) $y = x^2\sqrt{2x^2 \quad 4}$

(c) $y = 2x\sqrt{2x^2 \quad 2}$

(d) $y = x\sqrt{2x^2 \quad 4}$

(e) None of the above

15. Solve the particular solution of the differential equation:

$$y'' + y = \tan x, \quad 0 < x < \frac{\pi}{2}$$

(a) $y = (-\ln|\sec(x) + \tan(x)| + \sin(x))\cos(x) - \cos(x)\sin(x)$

(b) $y = (-\ln|\sec(x) + \tan(x)| + \sin(x))\cot(x) - \cos(x)\sin(x)$

(c) $y = (-\ln|\sin(x) + \tan(x)| + \sin(x))\cos(x) - \cos(x)\sin(x)$

(d) $y = (-\ln|\sec(x) + \tan(x)| + \sin(x))\cos(x) - \cos(x)\sec(x)$

(e) None of the above

16. Solve the differential equation:

$$x^2y'' + xy' - 9y = 0, \quad y_1(x) = x^3$$

(a) $y = cx^3 - \frac{1}{6}c_2x^{-3}$

(b) $y = cx^3 + \frac{1}{6}c_2x^{-3}$

(c) $y = -cx^3 - \frac{1}{6}c_2x^{-3}$

(d) $y = cx^3 - \frac{2}{6}c_2x^3$

(e) None of the above

17. Solve the differential equation:

$$xydx = xxdy$$

(a) $y'' = \frac{xx''}{y}$

(b) $y'' = \frac{yy''}{x}$

(c) $y'' = \frac{x^2y''}{y}$

(d) $y'' = \frac{y^2}{x}$

(e) None of the above

18. Solve the differential equation:

$$y'' - y' - 2y = 4x^2$$

(a) $y = c_1e^{2x} + c_2e^{-x} - 2x^2 + 2$

(b) $y = c_1e^{2x} + c_2e^{-x} - 2x^2 - 2$

(c) $y = c_1e^{2x} + c_2e^{-x} - 2x^2 + 2x$

(d) $y = c_1e^{2x} - c_2e^{-x} - 2x^2 + 2x$

(e) None of the above

19. Find the Wronskian of the functions $f(x)$ and $g(x)$

$$f(x) = 2x^2 \quad g(x) = x^4$$

(a) x^5

(b) $4x^5$

(c) 0

(d) $2x^5$

(e) None of the above

20. Which of the following represent the differential for of the equation below?

$$e^{y'+y} = x$$

(a) $dy - (\ln|x| + y)dx = 1$

(b) $dy - (\ln|x| - y)dx = 2$

(c) $dx - (\ln|x| + y)dx = 0$

(d) $dy - (\ln|x| - y)dx = 0$

(e) None of the above

21. Solve the differential equation given below:

$$9y''' + 12y'' + 4y' = 0$$

(a) $y = c_1 + c_2e^{-\frac{2}{3}x} + c_3xe^{-\frac{2}{3}x}$

(b) $y = c_1 + c_2e^{-\frac{2}{3}x} - c_3xe^{-\frac{2}{3}x}$

(c) $y = c_1 - c_2e^{\frac{2}{3}x} - c_3xe^{-\frac{2}{3}x}$

(d) $y = c_1 + c_2e^{-\frac{2}{3}x} + c_3xe^{-\frac{2}{3}x}$

(e) None of the above

22. Solve for the particular intergral:

$$y'' - 5y' + 6y = e^{2x}$$

(a) $y_p = -e^{2x} - xe^{4x}$

(b) $y_p = -e^{2x} - xe^{2x}$

(c) $y_p = -e^{4x} - xe^{2x}$

(d) $y_p = -e^{2x} - xe^{3x}$

(e) None of the above

23. Solve the differential equation:

$$\frac{d^3y}{dx^3} + \frac{d^2y}{dx^2} = 3e^x + 4x^2$$

(a) $y = c_1e^x + c_2x^2 + c_3e^{-x} + \frac{3}{2}e^{2x}$

(b) $y = c_1 + c_2x + c_3e^{-x} + \frac{3}{2}e^{2x}$

(c) $y = c_1e^x + c_2x^2 + c_3e^{-x} + \frac{3}{2}e^x$

(d) $y = c_1 + c_2x + c_3e^{-x} + \frac{3}{2}e^x$

(e) None of the above

24. Solve the differential equation:

$$(D+3)^2y = xe^{-3x}$$

(a) $y = e^{5x}(\frac{x^3}{6} + c_1x + c_2)$

(b) $y = e^{3x}(\frac{x^3}{6} + c_1x + c_2e^x)$

(c) $y = e^{3x}(\frac{x^3}{6} + c_1e^x + c_2)$

(d) $y = e^{3x}(\frac{x^3}{6} + c_1x + c_2)$

(e) None of the above

25. Solve the differential equation given below:

$$y'' + 4y' + 4y = 0, \quad y(0) = 1, \quad y'(0) = -1$$

(a) $y = e^{-2x}(1 + 4x)$

(b) $y = e^{-2x}(1 + 3x)$

(c) $y = e^{2x}(1 + 3x)$

(d) $y = e^{-2x}(1 - 3x)$

(e) None of the above

KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI

COLLEGE OF ENGINNERING

MID- SEMESTER EXAMINATIONS, 2012/2013

B.SC. ENGINEERING

MATH 251: MATHEMATICS III

YEAR TWO

TIME 1 HOUR

NOVEMBER, 2012

INDEX NUMBER ..... PROGRAM .....

Answer all questions by circling the correct answers on the question paper and shade the letter corresponding to the correct answer on the scannable sheet provided. Use supplementary sheets for rough work. Do not take any sheet out of the examination hall.

1. Consider the following autonomous equation $\frac{dy}{dt} = y^3 - y$. Find $\lim_{t\to\infty} y(t)$ provided $y(0) = 0.5$.

   (a) $+\infty$
   (b) 1
   (c) 0
   (d) -1
   (e) None of the above

2. The interval, where the solution to the initial value problem $(\ln t)y' + \frac{1}{t-3}y = t,\ y(2) = 0$ exist and is unique, is

   (a) $(1, 3)$
   (b) $(3, +\infty)$
   (c) $(0, 3)$
   (d) $(-\infty, 3)$
   (e) None of the above

3. Which of the following equations is a third order nonlinear O.D.E.:

   (a) $\frac{dy}{dt} = y + 1$

   (b) $\frac{d^3y}{dt^3} + \frac{d^2y}{dt^2} + \frac{dy}{dt} + y = 1$

   (c) $\frac{d^3y}{dt^3} + t^3y = 0$

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(d) $\frac{d^3y}{dt^3} + y\frac{dy}{dt} = 1$

(e) None of the above

4. Given differential equation $y' - 2y = e^{2t}$ with initial condition $y(0) = 2$. Which of the following is correct solution to this problem:

(a) $y(t) = e^{2t} + e^{-2t}$

(b) $y(t) = (t+2)e^{2t}$

(c) $y(t) = (t+2)e^{-2t}$

(d) $y(t) = 2e^{2t} - e^{-2t}$

(e) None of the above

5. Which of the following ordinary differential equations is linear of the second order?

(a) $y'' + 4yy' = 0$

(b) $t^2y'' + ty' - y = \ln t$

(c) $(y')^2 + 2y = 0$

(d) $t^2 + y' = e^{-t}$

(e) None of the above

6. Which of the following will be an integrating factor for this differential?
$ty' - 2y = 2\cos 2t$.

(a) $\frac{1}{t^2}$

(b) e^{-2t}

(c) $-t^2$

(d) $-2t$

(e) None of the above

7. Find the maximal interval in which the existence and uniqueness of the solution to th following initial value problem is guaranteed.

$$(\sin t)y' + \frac{1}{t-1}y = \frac{t}{t+1}, \quad y(2) = 3$$

(a) $(0, \pi)$

(b) $(0, 2)$

(c) $(-1, \pi)$

(d) $(1, \pi)$

(e) None of the above

8. Consider the following two pairs of solutions.

$$y_1(t) = e^t \quad and \quad y_2(t) = e^{t+1}$$

$$z_1(t) = \sin t \quad and \quad z_2(t) = \sin(2t)$$

(a) Both of these pairs are fundamental solutions.
(b) Only the second pair is a fundamental solution.
(c) Only the first pair is a fundamental solution.
(d) Neither pairs are fundamental solutions.
(e) None of the above

9. Consider all the solutions of the equation $y' + y = 10$. As $t \to \infty$, they will

(a) all approach $-\infty$
(b) all approach 0
(c) all approach 10
(d) some approach $+\infty$, some approach $-\infty$.
(e) None of the above

10. Suppose $y_1(t) = t$ and $y_2(t) = t^2$ are both solutions of the second order linear equation $y'' + p(t)y' + q(t)y = 0$.

All of the functions below are also solutions of the same equation, EXCEPT

(a) $y = t^2 - 1$
(b) $y = 5t^2$
(c) $y = -9t^2 + 17t$
(d) $y = 0$
(e) None of the above

11. Let $y_1(t)$ and $y_2(t)$ be any two solutions of the second order linear equation

$$(t-5)y'' + 6y' + \sec^2(t)y = 0$$

What is the general form of their Wronskian, $W(y_1, y_2)(t)$?

(a) Ce^{6t}
(b) Ce^{-6t}
(c) $C(t-5)^6$
(d) $\dfrac{C}{(t-5)^6}$
(e) None of the above

12. Find the general solution of $y''+2y'+17y=0$.

(a) $y=C_1e^{t}\cos(4t)+C_2e^{t}\sin(4t)$

(b) $y=C_1e^{-t}\cos(4t)+C_2e^{-t}\sin(4t)$

(c) $y=C_1e^{t}\cos(8t)+C_2e^{t}\sin(8t)$

(d) $y=C_1e^{-t}\cos(8t)+C_2e^{-t}\sin(8t)$

(e) None of the above

13. Let $y(t)$ be the solution of the initial value problem

$y''+y'-6y=0, \quad y(0)=7, \quad y'(0)=\lambda.$

Suppose $\lim_{t\to\infty} y(t)=0$, find the value of λ.

(a) 14

(b) 0

(c) -7

(d) -21

(e) None of the above

14. A suitable form for a particular solution $Y(t)$ to the differential equation

$y''-4y=e^{2t}\cos t$ is:

(a) $Y(t)=Ae^{2t}\cos t+Be^{2t}\sin t;$

(b) $Y(t)=Ae^{2t}\cos t$

(c) $Y(t)=Ate^{2t}\cos t+Bte^{2t}\sin t;$

(d) $Y(t)=Ate^{2t}\cos t$

(e) None of the above

15. The definite integral $\int_0^\infty te^{(4-s)t}\,dt$ represents the Laplace transform of which functio

below?

(a) te^{4-t}

(b) t^4e^{-t}

(c) te^{4t}

(d) te^{-4t}

(e) None of the above

16. Let $f(t)$ be the inverse Laplace transform of $F(s)=\dfrac{2s-1}{s^2+4s+5}$. Evaluate $f(\frac{\pi}{2})$.

(a) $-5e^{-\pi}$
(b) $-e^{-\pi}$
(c) $-2e^{-\pi}$
(d) $4e^{-\pi}$
(e) None of the above

17. Find the inverse Laplace transform of $F(s)=\dfrac{2s^2+20s+36}{s^3+6s^2+9s}$.

(a) $4-4e^{-3t}-2te^{-3t}$
(b) $4-2e^{3t}+38te^{3t}$
(c) $4+2e^{3t}-te^{3t}$
(d) $4-2e^{-3t}+2te^{-3t}$
(e) None of the above

18. Suppose $y(t)$ is the solution of the first order linear initial value problem $y'-5y=t^3e^{2t}$, $y(0)=4$. What is $Y(s)$, the Laplace transform of $y(t)$?

(a) $Y(s)=\dfrac{6}{(s-5)(s-2)^4}+\dfrac{4}{s-5}$
(b) $Y(s)=\dfrac{6}{(s-5)(s-2)^4}-\dfrac{4}{s-5}$
(c) $Y(s)=\dfrac{6}{(s-5)(s+2)^4}+\dfrac{4}{s-5}$
(d) $Y(s)=\dfrac{6}{(s-5)(s+2)^4}-\dfrac{4}{s-5}$
(e) None of the above

19. Find the Laplace transform of $f(t)=2\sin 3t+3\cos 4t$:

(a) $\dfrac{6}{s^2+9}+\dfrac{3s}{s^2+16}$
(b) $\dfrac{6}{s^2+4}+\dfrac{2}{s^2+9}$
(c) $\dfrac{3s}{s^2+16}+\dfrac{2s}{s^2+9}$

(d) $\frac{2}{s^2+9}+\frac{3}{s^2+16}$

(e) None of the above

20. Consider the differential equation $4y''+4y'+cy=0$. Find the value of c so that the function $y(t)=C_1e^{-\frac{t}{2}}+C_2te^{-\frac{t}{2}}$, where C_1 and C_2 are constants, is a solution of the equation.

(a) 2

(b) 1

(c) $\frac{1}{2}$

(d) $-\frac{1}{4}$

(e) None of the above

21. The improper integral $\frac{2}{3}\int_0^\infty \frac{tdt}{2e^{(s-4)t}}$ represents the Laplace transform of which function

(a) $\frac{1}{3}te^{-4t}$

(b) te^{4t}

(c) te^{-4t}

(d) $\frac{1}{3}te^{4t}$

(e) None of the above

22. If the Laplace transform of f is $F(s)=\frac{s-1}{s^2-s-2}$ and $f(0)=-1$, $f'(0)=3$, then the Laplace transform of f'' is

(a) $\frac{s-1}{s^2-s-2}-s-3$

(b) $\frac{s^3-s^2}{s^2-s-2}+s-3$

(c) $\frac{s^2-s}{s^2-s-2}-s-3$

(d) $\frac{s^3-s^2}{s^2-s-2}+s+3$

(e) None of the above

23. The inverse Laplace transform of $F(s)=\dfrac{s-1}{s^2+3s+2}$ is

(a) $-2e^{-t}+3e^{-2t}$

(b) $-e^{-t}+3e^{-2t}$

(c) $2e^{-t}-3e^{-2t}$

(d) $-2e^{-t}+3e^{-2t}$

(e) None of the above

24. Which of the following is the solution of the IVP $9y''+6y'+y=0, \quad y(1)=0, \quad y'(1)=1$?

(a) $y(t)=e^{1/3}e^{-t/3}-e^{1/3}te^{t/3}$

(b) $y(t)=te^{-t/3}$

(c) $y(t)=t^3/3-1/3$

(d) $y(t)=(t-1)e^{-(t-1)/3}$

(e) None of the above

25. Using the method of undetermined coefficients, a particular solution of $y''+y=e^{3t}-4t\cos 5t$ has the form.

(a) $Ae^{3t}+(Bt+C)\cos 5t$

(b) $Ae^{3t}+Bt\cos 5t+Ct\sin 5t$

(c) $Ae^{3t}+(Bt+C)\cos 5t+(Dt+E)\sin 5t$

(d) $Ae^{3t}+Bt\cos 5t$

(e) None of the above

26. Which of the following improper integrals defines the Laplace transform of $f(t)=\begin{cases}0 & \text{if } 0\le t\le 4\\ e^{2t} & \text{if } 4\le t\end{cases}$?

(a) $\int_0^\infty e^{-(s+2)t}\,dt$

(b) $\int_4^\infty e^{-(s-2)t}\,dt$

(c) $\int_0^\infty e^{-(s+2)t}\,dt$

(d) $\int_4^\infty e^{-(s-2)t}\,dt$

(e) None of the above

27. Which property of the Laplace transform is false?

(a) $L[f(t)-g(t)]=L[f(t)]-L[g(t)]$

(b) $L[f(t)g(t)] = L[f(t)]L[g(t)]$

(c) $L[cf(t)] = cL[f(t)]$, for any constant c.

(d) $L[f'''(t)] = s^3L[f(t)] - s^2f(0) - sf'(0) - f''(0)$

(e) None of the above

28. Suppose $y(t)$ is the solution of the first order linear initial value problem $y' - y = e^{-5t}$, $y(0) = 2$. What is the Laplace transform of $y(t)$?

(a) $F(s) = \dfrac{3}{(s-5)(s-1)}$

(b) $F(s) = \dfrac{2s+11}{(s+5)(s-1)}$

(c) $F(s) = \dfrac{-2s-9}{(s-5)(s-1)}$

(d) $F(s) = \dfrac{2s-9}{(s+5)(s-1)}$

(e) None of the above

29. Using the method of undetermined coefficients, the general solution of the differe equation, $y'' + 5y' + 6y = t^2 + 1$ has the form:

(a) $At^2 + Bt + C$

(b) $c_1e^{-2t} + c_2e^{-3t} + At^2 + Bt + C$

(c) $c_1e^{2t} + c_2e^{3t} + At^2 + Bt + C$

(d) $c_1e^{-2t} + c_2e^{-3t}$

(e) None of the above

30. Which of the following is the solution of the IVP $y'' + 8y' + 16y = 0$, $y(0) = 1$, $y'(0) = 0$?

(a) $y(t) = e^{-4t}$

(b) $y(t) = te^{-4t}$

(c) $y(t) = e^{-4t} + 4te^{-4t}$

(d) $y(t) = e^{4t} - 4te^{4t}$

(e) None of the above

KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI

COLLEGE OF ENGINNERING

END OF SEMESTER EXAMINATIONS, 2012/2013

B.SC. ENGINEERING

MATH 251: MATHEMATICS III

YEAR TWO

NOVEMBER, 2012 TIME 2 ½ HOURS

INDEX NUMBER .....PROGRAM.....

Answer all questions by circling the correct answers on the question paper and shade the letter corresponding to the correct answer on the scannable sheet provided. Use supplementary sheets for rough work. Do not take any sheet out of the examination hall.

1. Which of the following equations is exact?

   (a) $te^{y'}\frac{dy}{dt}+ye^{y'}+2t=0$

   (b) $\left(ye^{y'}+2t\right)\frac{dy}{dt}=te^{y'}$

   (c) $\left(ye^{y'}+2t\right)\frac{dy}{dt}+te^{y'}=0$

   (d) $te^{y'}\frac{dy}{dt}+ye^{y'}-2t=0$

   (e) None of the above

2. Find the general solution to the following system $X'(t)=\begin{bmatrix}0 & 1\\ -1 & -2\end{bmatrix}X$.

   (a) $X(t)=c_1\begin{bmatrix}1\\-1\end{bmatrix}e^{t}+c_2\left(\begin{bmatrix}1\\-1\end{bmatrix}te^{-t}+\begin{bmatrix}0\\1\end{bmatrix}e^{-t}\right)$

   (b) $X(t)=c_1\begin{bmatrix}1\\-1\end{bmatrix}e^{t}+c_2\left(\begin{bmatrix}1\\-1\end{bmatrix}te^{-t}+\begin{bmatrix}0\\1\end{bmatrix}e^{-t}\right)$

   (c) $X(t)=c_1\begin{bmatrix}1\\-1\end{bmatrix}e^{-t}+c_2\left(\begin{bmatrix}1\\-1\end{bmatrix}te^{-t}+\begin{bmatrix}0\\1\end{bmatrix}e^{-t}\right)$

   (d) $X(t)=c_1\begin{bmatrix}1\\-1\end{bmatrix}e^{t}+c_2\left(\begin{bmatrix}1\\-1\end{bmatrix}t^2e^{t}+\begin{bmatrix}0\\1\end{bmatrix}e^{t}\right)$

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(e) None of the above

3. Using Q2, find the specific solution which meets the following initial condition $X(0)=\begin{bmatrix}2\\1\end{bmatrix}$.

(a) $X(t)=-\begin{bmatrix}1\\-1\end{bmatrix}e^{t}+3\left(\begin{bmatrix}1\\-1\end{bmatrix}te^{-t}+\begin{bmatrix}0\\1\end{bmatrix}e^{-t}\right)$

(b) $X(t)=-\begin{bmatrix}1\\-1\end{bmatrix}e^{t}+3\left(\begin{bmatrix}1\\-1\end{bmatrix}te^{-t}+\begin{bmatrix}0\\1\end{bmatrix}e^{-t}\right)$

(c) $X(t)=-\begin{bmatrix}1\\-1\end{bmatrix}e^{-t}+3\left(\begin{bmatrix}1\\-1\end{bmatrix}te^{-t}+\begin{bmatrix}0\\1\end{bmatrix}e^{-t}\right)$

(d) $X(t)=-\begin{bmatrix}1\\-1\end{bmatrix}e^{t}+3\left(\begin{bmatrix}1\\-1\end{bmatrix}t^{2}e^{t}+\begin{bmatrix}0\\1\end{bmatrix}e^{t}\right)$

(e) None of the above

4. Find the value a for which the given equation $\left(xy^{2}+ax^{2}y\right)+(x+y)x^{2}\frac{dy}{dx}=0$ is exa

(a) $a=1$
(b) $a=2$
(c) $a=3$
(d) $a=4$
(e) None of the above

5. Consider the Fourier series of the function given below.

$$f(x)=\begin{cases}x^{2}+1 & -2<x<\frac{1}{2}\\ \sin(2\pi x) & \frac{1}{2}\le x<2\end{cases},\quad f(x+4)=f(x)$$

To what value does the Fourier series converges to at $x=\frac{1}{2}$?

(a) 5/4
(b) 5/8
(c) 0
(d) 1/2
(e) None of the above

(c) $xy\dfrac{dy}{dx} = x^3 + y^3$

(d) $(x+y)y' = x - y$

(e) None of the above

10. One of the following equations is exact. Identify which one, and find a solution y of that equation satisfying $y(0) = 0$. what is $y(1)$?

$(2x+2y) + (2x+2y)\dfrac{dy}{dx} = 0,$ $\qquad$ $(2x+3y) + (2x+3y)\dfrac{dy}{dx} = 0.$

(a) -2

(b) -1

(c) 0

(d) 1

(e) None of the above

11. At any time t, the rate of growth of the population N of elk in Lone Elk Park is proportional to the product of N and $L - N$, where $L = 500$ is the maximum number the park can maintain. At $t = 0$ there are 100 elk in the park, and at $t = 4$ there are 2 in the park. Find the number of elk when $t = 1$?

(a) 100

(b) 120

(c) 121

(d) 150

(e) None of the above

12. Find the characteristic equation for the equation $y^{(iv)} + 3y'' - y' + 2y = x$.

(a) $r^4 + 3r^3 - r + 2 = 0$

(b) $r^4 + 3r^3 - r + 2 = x$

(c) $r^4 + 3r^2 - r + 2 = 0$

(d) $r^4 + 3r^2 - r + 2 = x$

(e) None of the above

13. Suppose a certain population satisfies the initial value problem

$$\frac{dP}{dt} = 50P(600 - P), \quad P(0) = 2000.$$

Choose the statement that correctly describes the concavity of the solution curve.

(a) The graph is concave up everywhere

(b) The graph is concave down everywhere
(c) There is an inflection point where $P = 2000$
(d) There is an inflection point where $P = 3000$
(e) None of the above

14. An arrow is shot straight upward from the ground with initial velocity of 160 ft/s. It experiences both the deceleration of gravity and deceleration $v^2/800$ due to air resistance. How high in the air does it go?
(a) 273 ft
(b) 275 ft
(c) 277 ft
(d) 279 ft
(e) None of the above

15. It is known that the matrix $A = \begin{bmatrix} 2 & 1 \\ a & 2 \end{bmatrix}$ has eigenvalues $\lambda = 1$ and $\lambda = 3$. What is a?

(a) -1
(b) 0
(c) 1
(d) 2
(e) None of the above

16. Find the solution of the initial value problem $x' = \begin{bmatrix} 4 & 2 \\ 3 & -1 \end{bmatrix} x, \quad x(0) = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

(a) $\frac{1}{7}\begin{bmatrix} 3e^{-2t} + 4e^{5t} \\ -9e^{-2t} + 2e^{5t} \end{bmatrix}$
(b) $\frac{1}{7}\begin{bmatrix} 4e^{-2t} + 3e^{5t} \\ 2e^{-2t} - 9e^{5t} \end{bmatrix}$
(c) $\frac{1}{7}\begin{bmatrix} 9e^{-2t} - 2e^{5t} \\ -4e^{-2t} - 3e^{5t} \end{bmatrix}$
(d) $\frac{1}{7}\begin{bmatrix} 3e^{2t} + 4e^{-5t} \\ -9e^{2t} + 2e^{-5t} \end{bmatrix}$
(e) None of the above

17. Solve $x' = \begin{bmatrix} 2 & 2 \\ 0 & 2 \end{bmatrix} x + \begin{bmatrix} 4 \\ e^{2t} \end{bmatrix}$.

(a) $\begin{bmatrix} -2+t^2e^{2t} \\ e^{2t} \end{bmatrix}$

(b) $\begin{bmatrix} t^2e^{2t} \\ e^{2t} \end{bmatrix}$

(c) $\begin{bmatrix} 2+t^2e^{2t} \\ e^{2t} \end{bmatrix}$

(d) $\begin{bmatrix} -2+t^2e^{2t} \\ te^{2t} \end{bmatrix}$

(e) None of the above

18. Let $F(s)$ be the Laplace transform of the function $f(t)=\begin{cases} 0, & t<1 \\ t^2, & t\geq 1. \end{cases}$ Find F

(a) e^{-1}

(b) 5

(c) $5e^{-1}$

(d) e

(e) None of the above

19. Let g be the inverse Laplace transform of $G(s)=\frac{1}{s}-\frac{s+2}{s^2+4}$. What is $g(1)$? Choo

closest value.

(a) -0.9

(b) -0.2

(c) 0.5

(d) 1.2

(e) None of the above

20. Suppose that in the absence of other factors a population of fish in a certain lake i at a rate proportional to the size of the population. However, each day a fisherman catches 10 out of the lake. If $P(t)$ is the number of fish after t days, which of the following differential equations models this situation?

(a) $\frac{dP}{dt}=-10k$ for some $k>0$

(b) $\frac{dP}{dt}=10-kP$ for some $k>0$

(c) $\frac{dP}{dt} = 10kP$ for some $k > 0$

(d) $\frac{dP}{dt} = kP - 10$ for some $k > 0$

(e) None of the above

21. Find the general solution to $xy^2 \frac{dy}{dx} = y^3 + x^4$. *Hint:* Use the substitution $u = \frac{y}{x}$

(a) $y = c(x^4 + y^4)$

(b) $y = \sqrt[3]{3x + c}$

(c) $y = \frac{1}{x}\sqrt[3]{3x + c}$

(d) $y = x\sqrt[3]{3x + c}$

(e) None of the above

22. Which of the following is general solution (given implicitly) to the differential equation $(3x^2y - 3y)dx + (x^3 - 3x + 2y)dy = 0$?

(a) $x^3 - xy^3 = c$

(b) $x^3y - 3xy + y^2 = c$

(c) $x^3 + xy^3 = c$

(d) $3x^2 - 3 = c$

(e) None of the above

23. Transform the following system into a system of first-order differential equations $\begin{matrix} x'' - 5x + 4y = 0 \\ y'' + 4x - 5y = 0 \end{matrix}$

(a) $\begin{matrix} x_1' = 5x_1 - 4x_2 \\ x_2' = -4x_1 + 5x_2 \\ x_3' = 0 \\ x_4' = 0 \end{matrix}$

(b) $\begin{matrix} x_1' = 5x_2 - 4x_3 \\ x_2' = -4x_3 + 5x_4 \\ x_3' = x_4 \\ x_4' = 0 \end{matrix}$

(c)
$$\begin{aligned} x_1' &= x_2 \\ x_2' &= 5x_1 - 4x_2 \\ x_3' &= x_4 \\ x_4' &= -4x_1 + 5x_3 \end{aligned}$$

(d)
$$\begin{aligned} x_1' &= x_2 \\ x_2' &= 5x_1 - 4x_3 \\ x_3' &= x_4 \\ x_4' &= -4x_1 + 5x_3 \end{aligned}$$

(e) None of the above

24. Use Laplace transform to rewrite $2y''+y'+7y=0,\ y(0)=4,\ y'(0)=8$ as an alge equation.

(a) $Y(s)=\dfrac{2s^2+s+7}{8s+20}$

(b) $Y(s)=2s^2+s+7$

(c) $Y(s)=\dfrac{8s+20}{2s^2+s+7}$

(d) $Y(s)=\dfrac{7s^2-s+2}{3s-5}$

(e) None of the above

25. What is the form of a particular solution of $x'=\begin{bmatrix}1 & 5\\ 2 & 4\end{bmatrix}x+\begin{bmatrix}1\\ te^{-t}\end{bmatrix}$? Below, a, b, c denote vector coefficients.

(a) $ae^{-t}+bte^{-t}$

(b) $ate^{-t}+bt^2e^{-t}$

(c) $ae^{-t}+bte^{-t}+ct^2e^{-t}$

(d) $a+be^{-t}+cte^{-t}+dt^2e^{-t}$

(e) None of the above

26. Let $g(t)$ be the inverse Laplace transform of $G(s)=\dfrac{2s-3}{s^2+9}$. Find $g(0)$.

(a) -2

(b) -1

(c) 0

(d) 2

(e) None of the above

27. Suppose that $f(t)=\begin{cases}0 & t<1\\ t^2 & t\geq 1\end{cases}$. If F is the Laplace transform of f, what is $e^2F(2)$?

(a) $\frac{1}{2}$

(b) $\frac{3}{2}$

(c) $\frac{1}{4}$

(d) $\frac{5}{4}$

(e) None of the above

28. Solve the initial value problem $x'=\begin{bmatrix}1 & 3\\ 0 & 1\end{bmatrix}x+\begin{bmatrix}2e^t\\ 2te^t\end{bmatrix}$, $x(0)=\begin{bmatrix}0\\ 0\end{bmatrix}$.

(a) $\begin{bmatrix}2t+t^3\\ t^2\end{bmatrix}e^t$

(b) $\begin{bmatrix}t^2\\ 2t+t^3\end{bmatrix}e^t$

(c) $\begin{bmatrix}t+3t^2\\ -4t+t^2\end{bmatrix}e^t$

(d) $\begin{bmatrix}-4t+t^2\\ t+3t^2\end{bmatrix}e^t$

(e) None of the above

29. Use "substitution" to find a solution of the initial value problem

$xy'=y+\sqrt{x^2-y^2}$, $y(1)=0$ for $x>0$. *Hint:* One or several of the following integrals may be useful, $\int\frac{du}{\sqrt{1-u^2}}=\arcsin u+C$, $\int\frac{du}{1-u^2}=\frac{1}{2}\ln\left|\frac{u+1}{u-1}\right|+C$.

(a) $y=x\sin(\ln x)$

(d) $y = \cos(\ln x)$

(e) None of the above

30. Solve $\frac{dy}{dx} + y = \frac{\cos x}{e^x}$, $y(0) = 1$. What is $y(\pi)$?

(a) $e^{\pi} + 1$

(b) e^{π}

(c) $\frac{2}{e^{\pi}}$

(d) $\frac{1}{e^{\pi}} + 1$

(e) $\frac{1}{e^{\pi}}$

31. Find a second order linear differential equation with constant coefficients to which the following functions are solutions $y_1(t) = e^{-2t}$, $y_2(t) = e^{t}$.

(a) $y'' - y' - 2y = 0$

(b) $y'' + 2y' - 3y = 0$

(c) $y'' + y' - 2y = 0$

(d) $y'' + y' + 2y = 0$

(e) None of the above

32. Find the general solution of $y' - 6y = te^{6t}$.

(a) $y = \frac{1}{12} te^{12t} + Ce^{6t}$

(b) $y = e^{-6t} + C$

(c) $y = te^{-6t} + Ce^{-6t}$

(d) $y = \frac{1}{2} t^2 e^{6t} + Ce^{6t}$

(e) None of the above

33. Which pair of functions below cannot be a fundamental set of solutions?

(a) e^{5t}, $2e^{-5t}$

(b) $\cos t$, $-9\sin t$

(c) $2e^{-t}$, $4e^{-t+3}$

(d) $e^{2t}, \quad e^{2t}-6$

(e) None of the above

34. Which of the following second order homogeneous linear equations below does not have $y = e^{-4t}$ as a solution?

(a) $2y'' + 8y' = 0$

(b) $y'' - 2y' - 8y = 0$

(c) $y'' - 16y = 0$

(d) $-y'' - 5y' - 4y = 0$

(e) None of the above

35. Which of the following is an integrating factor for the differential equation : $ty' - (t+2)y = 2\tan(3t)$.

(a) $\mu - \frac{1}{t^2 e^t} = 0$

(b) $\mu = \frac{t+2}{t}$

(c) $\mu - \frac{1}{t^2} = 0$

(d) $\mu = t^2 e^t$

(e) None of the above

36. The solution of the initial value problem $\frac{dy}{dx} = y^{-1}(1-x), \quad y(-1) = 1$ is defined for

(a) $x < 0$

(b) $x < 1 + \sqrt{5}$

(c) $|x-1| < \sqrt{5}$

(d) all real x

(e) None of the above

37. Let $y(t)$ be the solution of the initial value problem $y'' + y' - 2y = 0, \quad y(0) = b, \quad y'(0) = 2$. For which value of b is $\lim_{t \to +\infty} y(t) = 0$?

(a) there is no value for b with the limit 0

(b) 0

(c) -1

(d) 1

(e) None of the above

38. For which value of λ are the functions $f_1(t) = t^2$ and $f_2(t) = t^2 + \lambda t^3$ linearly dependent?

(a) $\lambda = -1$

(b) $\lambda = 0$

(c) $\lambda = 1$

(d) $\lambda = 2$

(e) None of the above

39. Let y_1 and y_2 be two independent solutions of the differential equation $ty'' + y' +$ Then the Wronskian $W(y_1, y_2)(t)$ of the equation equals (with c a constant).

(a) ce^{-t}

(b) ce^{t}

(c) ct^{-1}

(d) ct

(e) None of the above

40. Suppose y_1 and y_2 are two solutions of the second order linear equation $t^2y'' + 2ty' - 5e^{3t}y = 0$. What is their Wronskian $W(y_1, y_2)(t)$ as a function of time if $W(y_1, y_2)(1) = 3$?

(a) $W(y_1, y_2)(t) = 3t^2$

(b) $W(y_1, y_2)(t) = 3e^{-t^2}$

(c) $W(y_1, y_2)(t) = \frac{3}{t^2}$

(d) $W(y_1, y_2)(t) = 3e^{t^2-1}$

(e) None of the above

41. Suppose $y_1 = 2e^{2t} + te^{2t}$ and $y_2 = -e^{5t} + te^{2t}$ are both solutions of the second order equation $y'' + p(t)y' + q(t)y = g(t)$. Which of the functions below is also a soluti the same equation?

(a) $y = 2e^{2t} - e^{5t} + 2te^{2t}$

(b) $y = -2e^{2t} - te^{2t}$

(c) $y = 2e^{5t} + te^{2t}$

(d) $y = 2e^{2t} + e^{5t}$

(e) None of the above

42. Which of the following is the general solution of $4y'' + y = 0$?

(a) $y(t) = c_1 e^{t} + c_2 e^{-t}$

(b) $y(t) = c_1 \cos(t/2) + c_2 \sin(t/2)$

(c) $y(t) = (c_1 + c_2 t) e^{t/2}$

(d) $y(t) = e^{-t}(c_1 \cos(t) + c_2 \sin(t))$

(e) None of the above

43. Which of the following could be a particular solution to $y + 4y + 4y = te^{t} + \cos(2t) + t + 2$, for some constants $A_1, A_2, \ldots, A_6$?

(a) $y(t) = (A_1 + A_2 t) te^{t} + A_3 \sin(2t) + A_4 \cos(2t) + A_5 t + A_6$

(b) $y(t) = (A_1 + A_2 t) e^{t} + A_3 \sin(t) + A_4 \cos(t) + A_5 t + A_6$

(c) $y(t) = (A_1 + A_2 t) e^{t} + A_3 t \sin(2t) + A_4 t \cos 2(t) + A_5 t + A_6$

(d) $y(t) = (A_1 + A_2 t) e^{t} + A_3 \sin(2t) + A_4 \cos(2t) + A_5 t + A_6$

(e) None of the above

44. Which of the following pairs of functions are linearly independent?

(a) $f(t) = t^3, \quad g(t) = 2t^3$

(b) $f(t) = \sin^2(t), \quad g(t) = \cos(2t) - 1$

(c) $f(t) = e^{t+2}, \quad g(t) = e^{t}$

(d) $f(t) = \sin(t), \quad g(t) = \cos(t)$

(e) None of the above

45. The Laplace transform of $e^{t} + te^{2t} + e^{3t}\cos(4t)$ is

(a) $\dfrac{1}{s+1} + \dfrac{1}{(s+2)^2} + \dfrac{s+3}{(s+3)^2 + 16}$

(b) $e^{s}\dfrac{1}{s} + e^{s}\dfrac{1}{s^2} + e^{s}\dfrac{s}{s^2 + 16}$

(c) $\dfrac{1}{s-1}+\dfrac{1}{(s-2)^2}+\dfrac{s+3}{(s-3)^2+16}$

(d) $e^{-s}\dfrac{1}{s}+e^{-s}\dfrac{1}{s^2}+e^{-s}\dfrac{s}{s^2+16}$

(e) None of the above

46. Given the eigenvalues and eigenvector of A as:

$\lambda_1 = 4+3i, v_1 = \begin{pmatrix} 1 \\ -i \end{pmatrix}$, $\lambda_2 = 4-3i, v_2 = \begin{pmatrix} 1 \\ i \end{pmatrix}$. Which of the following is the gener... solution to $X' = AX$:

(a) $C_1 e^{-4t}\begin{pmatrix} \cos 3t \\ \cos 3t + \sin 3t \end{pmatrix} + C_2 e^{-4t}\begin{pmatrix} \sin 3t \\ \sin 3t - \cos 3t \end{pmatrix}$

(b) $C_1\begin{pmatrix} 1 \\ 1 \end{pmatrix}e^{t} + C_2\begin{pmatrix} -1 \\ 1 \end{pmatrix}e^{7t}$

(c) $C_1 e^{-4t}\begin{pmatrix} \cos 6t \\ \sin 6t \end{pmatrix} + C_2 e^{-4t}\begin{pmatrix} \sin 6t \\ -\cos 6t \end{pmatrix}$

(d) $C_1 e^{4t}\begin{pmatrix} \cos 3t \\ \sin 3t \end{pmatrix} + C_2 e^{4t}\begin{pmatrix} \sin 3t \\ -\cos 3t \end{pmatrix}$

(e) None of the above

47. Consider the differential equation $y' = 2y - t^3$. Which statement below is true?

(a) It is a linear equation.
(b) It is a third order equation.
(c) It is an autonomous equation.
(d) It is a separable equation.
(e) None of the above

48. Consider the equation $y' = y(y-20)(y-100)$. Suppose $y(5) = \lambda$ and $\lim_{t\to\infty} y(t) =$ ... all possible value(s) of λ.

(a) $\lambda = 20$
(b) $0 < \lambda < 20$
(c) $0 < \lambda < 100$
(d) $20 < \lambda < 100$
(e) None of the above

49. Consider the first order linear equation $ty' - 10y = t^3 e^{3t}, \quad t > 0$. What is a suitable integrating factor that could be used to solve this equation?

(a) $\mu(t) = \frac{1}{t^{10}}$

(b) $\mu(t) = t^{10}$

(c) $\mu(t) = e^{10t}$

(d) $\mu(t) = e^{-10t}$

(e) None of the above

50. Consider all the solutions of the equation $y'' + 2y' + y = 0$. As $t \to \infty$ they will

(a) all approach $-\infty$ or $+\infty$.

(b) all approach 0.

(c) some approach 0, some approach $-\infty$, some approach $+\infty$.

(d) have no limit, since they are oscillating.

(e) None of the above

51. Consider the equation $y' = \frac{\sin 2t}{\cos y}$. Which function below is its general solution?

(a) $y = \sin^{-1}\left(-\frac{1}{2}\cos 2t + C\right)$

(b) $y = \sin^{-1}(-2\cos 2t) + C$

(c) $y = -\frac{1}{2}\cos 2t + C$

(d) $y = -2\ln(\cos y) + C$

(e) None of the above

52. Suppose it is known that $y = 2e^{\pi t}$ is a solution of $y'' + 4y' + 4y = g(t)$. All of the functions below are also solutions of the equation, EXCEPT

(a) $y = 2\pi e^{\pi t}$

(b) $y = 2e^{\pi t} - 2\pi e^{-2t}$

(c) $y = 9te^{-2t} + 2e^{\pi t}$

(d) $y = \frac{1}{\pi}e^{-2t} - 2\pi te^{-2t} + 2e^{\pi t}$

(e) None of the above

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53. What is the general solution of $y'' + 4y' + 3y = 4e^{-2t}$?

(a) $C_1e^{-t} + C_2e^{-3t} + 4te^{-2t}$

(b) $C_1e^{-t} + C_2e^{-3t} - 4e^{-2t}$

(c) $C_1e^{-t} + C_2e^{-3t} + 4e^{-2t}$

(d) $C_1e^{-t} + C_2e^{-3t} - 4te^{-2t}$

(e) None of the above

54. Consider the third order linear equation $y''' - 9y' = 0$. What is its general solution?

(a) $y(t) = C_1 + C_2 \cos 3t + C_3 \sin 3t$

(b) $y(t) = C_1 + C_2t + C_3e^{9t}$

(c) $y(t) = C_1 + C_2e^{3t} + C_3e^{-3t}$

(d) $y(t) = C_1 + C_2e^{3t} + C_3te^{3t}$

(e) None of the above

55. A 200-liter tank initially contains 40 liters of 3 grams/liter potassium chloride solu
At $t = 0$, potassium chloride solution with a concentration of 6 grams/liter begins to
into the tank at the rate of 4 liters/min. The thoroughly mixed content of the tank is
off at the rate of 2 liters/min. Which of the initial value problems below best descri
quantity of potassium chloride, $Q(t)$, that would be in the tank at time t, $0 <= t < 80$?

(a) $Q' = 24 - \frac{1}{20}Q, \quad Q(0) = 120$

(b) $Q' = 6 - \frac{1}{20+t}Q, \quad Q(0) = 600$

(c) $Q' = 6 - \frac{1}{20}Q, \quad Q(0) = 600$

(d) $Q' = 24 - \frac{1}{20+t}Q, \quad Q(0) = 120$

(e) None of the above

56. Which system of first order linear equations below is equivalent to the second orde
linear equation $y'' - 5y' + 3y = 0$?

(a) $\begin{cases} x_1' = & x_2 \\ x_2' = & 3x_1 - 5x_2 \end{cases}$

(b) $\begin{cases} x_1' = & x_2 \\ x_2' = & -3x_1 + 5x_2 \end{cases}$

(c) $\begin{cases} x_1' = & x_2 \\ x_2' = & -5x_1 + 3x_2 \end{cases}$

(d) $\begin{cases} x_1' = & x_2 \\ x_2' = & 5x_1 - 3x_2 \end{cases}$

(e) None of the above

57. The inverse Laplace transform of $F(s) = \dfrac{2s-1}{s^2+2s+5}$ is

(a) $2e^t \cos 2t - e^t \sin 2t$

(b) $2e^{-t} \cos 2t - e^t \sin 2t$

(c) $2e^{2t} \cos t - \dfrac{1}{2} e^{2t} \sin t$

(d) $2e^{-t} \cos 2t - \dfrac{3}{2} e^{-t} \sin 2t$

(e) None of the above

58. Which system of first order linear equations below is equivalent to the second order linear equation $y'' + y' - 2y = 0$?

(a) $\begin{cases} x_1' = & x_2 \\ x_2' = & -2x_1 + x_2 \end{cases}$

(b) $\begin{cases} x_1' = & x_1 \\ x_2' = & x_1 - 2x_2 \end{cases}$

(c) $\begin{cases} x_1' = & x_2 \\ x_2' = & 2x_1 - x_2 \end{cases}$

(d) $\begin{cases} x_1' = & -2x_1 + x_2 \\ x_2' = & x_1 \end{cases}$

(e) None of the above

59. Solve the initial value problem $x' = \begin{pmatrix} -1 & -1 \\ 1 & -1 \end{pmatrix} x, \quad x(0) = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$.

(a) $x = e^t \begin{pmatrix} 2\cos t + \sin t \\ -\cos t + 2\sin t \end{pmatrix}$

(b) $x = e^{2t}\begin{pmatrix} 2\cos t + \sin t \\ -\cos t + 2\sin t \end{pmatrix}$

(c) $x = e^{-t}\begin{pmatrix} 2\cos t + \sin t \\ -\cos t + 2\sin t \end{pmatrix}$

(d) $x = e^{2t}\begin{pmatrix} 2\cos t - \sin t \\ \cos t + 2\sin t \end{pmatrix}$

(e) None of the above

60. Consider the following two differential equations:

I. $y'' + ay' + by = 0 \quad y(0) = 0 \quad y'(0) = 2$

II. $y'' + ay' + by = 0 \quad y(0) = 0 \quad y(\pi) = 2$

Where $a, b \in R$ Which of the following statements are true?

(a) Both I and II always have a unique solution on some interval.

(b) Only I always has a solution on some interval.

(c) Only II always has a solution on some interval.

(d) None of the above

KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY

COLLEGE OF ENGINEERING

MATHEMATICS SECOND YEAR

SUPPLEMENTARY EXAMINATION AUGUST 2011

MATH 251 MATHEMATICS III (Differential Equations) TIME: 2 hours

Index Number: _____ **Dept.**_____

Instructions:

1. Check that you have 10 pages of questions.
2. Shade the correct answers on the scanable sheet provided and circle the correct answers on the question paper.
3. Submit both the scanable sheet and the question paper at the end of the examination.
4. In the event that your chosen answer is repeated in a question, shade the first of the options

Answer all questions

1. The differential equation $e^{2t}y'' - 2y' + 3y = \sqrt[3]{t}$ is
   a. linear of order 1
   b. linear of order 2
   c. nonlinear of order 1
   d. nonlinear of order 2
   e. none of the above

2. If $\frac{dy}{dx} = x^2 \ln(y), \quad y(1) = 4$

Then the value of the second derivative of $y(x)$ evaluated at $x = 1$ is

a. $y''(1) = 5\ln(2)$
b. $y''(1) = \frac{9}{2}\ln(2)$
c. $y''(1) = 4\ln(2)$
d. $y''(1) = \frac{7}{2}\ln(2)$
e. none of the above

3. An integrating factor for the following differential equation:
$(y+2x-1)dx + x(x+y)dy = 0$
is

a. $e^x y^2$
b. $e^x y$
c. e^x
d. y^2
e. none of the above

4. For solving the differential equation: $x^2y'' + xy' + y = 0$, the adequate substitution is

a. $y = (\ln x)'''$
b. $y = e^{\ln x}$
c. $y = \ln x$
d. $y = x'''$
e. none of the above

5. Determine the degree of the differential equation $\left[\frac{d^2y}{dx^2} + \frac{dy}{dx}\right]^{\frac{1}{3}} = \left[\frac{dy}{dx} + \sin^2 x\right]^{\frac{2}{3}}$

a. 1
b. 3
c. 5
d. 10
e. none of the above

6. The following differential equations are linear except

a. $y'' - 3y' + 2y = 0$
b. $\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} = 0$
c. $xdx + ydy = 0$
d. $L\frac{di}{dt} + Ri = E$
e. none of the above

7. Find the differential equation whose solution is $y = c_0 + c_1x + c_2x^2 + c_3x^3$.

a. $y^{(iv)} + y''' + y'' + y' + y = 0$
b. $y''' + y = 0$
c. $y^{(iv)} - y = 0$
d. $y^{(iv)} = 0$
e. none of the above

8. Determine the differential equation whose solution is $y = c_1x + c_2e^x$.

a. $(x-1)y'' - xy' + y = 0$
b. $xy'' - xy' - y = 0$
c. $xy'' + xy' - y = 0$
d. $(x-1)y'' + xy' - y = 0$
e. none of the above

9. Solve the differential equation $(1+x^2)-x(1+2y)y'=0; y(1)=0$.

a. $y=\ln x+\frac{x^2}{2}+y^2$

b. $y=\ln x+\frac{x^2}{2}+y^2-\frac{1}{2}$

c. $y=\frac{1}{x}+\frac{x^2}{2}+y^2-\frac{1}{2}$

d. $y=\frac{1}{x}+\frac{x^2}{2}+y^2$

e. none of the above

10. Solve the differential equation $(y\cos x+2xe^y)+(\sin x+x^2e^y)y'=0$.

a. $2y+y\sin x+x^2e^y=c$

b. $y\sin x+x^2e^y+2=c$

c. $y-y\sin x+x^2e^y=c$

d. $2y=y\sin x+x^2e^y+c$

e. none of the above

11. The differential equation $\left(\frac{\sin y}{y}-2e^{-x}\sin x\right)dx+\left(\frac{\cos y+2e^{-x}\cos x}{y}\right)dy=0$ has the integrating factor, $\mu(x,y)=ye^x$. Solve the differential equation.

a. $e^x\sin y+y\cos x=c$

b. $e^x\sin y+y^2\cos x=c$

c. $e^x\cos y+2y\cos x=c$

d. $e^x\sin y+2y\cos x=c$

e. none of the above

12. Find $f(x,y)=c$, which is defined such that $\frac{\partial f}{\partial x}=y^2(3x^2+y^2)$ and $\frac{\partial f}{\partial y}=2xy(x^2+2y^2)$.

a. $x^2y^3+xy^4=c$

b. $xy^2(x^2+y^2)=c$

c. $x^2y(x^2+y^2)=c$

d. $x^4y+x^2y^3=c$

e. none of the above

13. Solve the differential equation $\left(3x - 4y^3\right)y' + y = 0$.

a. $3y^3 = 2xy^6 + c$

b. $xy^3 = \frac{1}{3}y^6 + c$

c. $xy^3 = \frac{2}{3}y^6 + c$

d. $y^3 = \frac{2}{3}y^6 + c$

e. none of the above

14. Solve the differential equation $x^2 y - x^3 \frac{dy}{dx} = y^4 \cos x$.

a. $y^3\left(\sin x + c\right) = x^3$

b. $y^3x^3 = 3\sin x + c$

c. $y^3\left(3\sin x + c\right) = x^3$

d. $y^3 = 3x^3 \sin x + c$

e. none of the above

15. Which of the following second order homogeneous linear equations has $y_1(t) = e^{2t}$ as two of its solutions?

a. $2y'' + 2y' - 12y = 0$

b. $y'' + 5y' + 6y = 0$

c. $y'' - y' - 6y = 0$

d. $2y'' - 3y' = 0$

e. none of the above

16. All of the following pairs of functions are linearly independent on $(-\infty, \infty)$ except

a. $1, e^t$

b. $\cos t,\ \sin(t + 2\pi)$

c. $e^{-2t}\cos 2t,\ e^{-2t}\sin 2t$

d. $e^{2t},\ e^{5t-1}$

e. none of the above

17. Suppose $y_1 = x + 1$ and $y_2 = x^2$ are two solutions of a second order linear equation, what is their Wronskian $W(y_1, y_2)(t)$?

a. $2x$

b. $x^2 - 2x$

c. x^2

d. $x^2 + 2x$

e. none of the above

18. The equation $y = \sin(Ax + B)$ is the general solution of the differential equation

a. $(1 - y^2)y'' - y(y')^2 = 0$

b. $y^2y'' - (y')^2 = 0$

c. $(y'')^2 - y^2y' = 0$

d. $y^2y'' - (y'/y)^2 = 0$

e. none of the above

19. Find the general solution the differential equation: $y''' + y' = 1$.

a. $y = A + B\cos t + C\sin t + t/2$

b. $y = A + B\cos t + C\sin t + 1$

c. $y = A + B\cos t + C\sin t + t$

d. $y = At + B\cos t + C\sin t + 2$

e. none of the above

20. Which of the following is not an Autonomous system?

a. $x' + 3ty' + 2x + y = 0$
$x' - 2y - 3x - 5y = 0$

b. $x' = 3x + 2y$
$y' = 3x - 2y$

c. $x' = x + y$
$y' = x^2 - xy^3$

d. $x' = e^x + y$
$y' = \sin x - \log y$

e. none of the above

21. Which of the following is a general solution of the system:

$$(D-4)x + 3y = 0$$
$$-6x + (D+7)y = 0$$

a. $x = c_1e^{2t} + 3c_2e^{-5t}, y = c_1e^{2t} + 2c_2e^{-5t}$

b. $x = c_1e^{2t} - 3c_2e^{-5t}, y = c_1e^{2t} + 2c_2e^{-5t}$

c. $x = c_1e^{2t} + 3c_2e^{-5t}, y = c_1e^{2t} - 2c_2e^{-5t}$

d. $x = c_1e^{2t} + c_2e^{-5t}, y = c_1e^{2t} + c_2e^{-5t}$

e. none of the above

22. Find the solution of the system: $x' = 4x - 3y$, $y' = 6x - 7y$ that satisfy the initial conditions; $x(0) = 2$, $y(0) = -1$

a. $x = e^{2t} - e^{-5t}, y = e^{2t} - 3e^{-5t}$

b. $x = 3e^{2t} - e^{-5t}, y = e^{2t} - 3e^{-5t}$

c. $x = 3e^{2t} - e^{-5t}, y = 2e^{2t} - 3e^{-5t}$

d. $x = 3e^{2t} + e^{-5t}, y = 2e^{2t} + 3e^{-5t}$

e. none of the above

23. Which of the following is a general solution of the system:

$x_1' = 3x_1 - x_2, x_2' = x_1 + x_2$

a. $x_1 = (c_1 + c_2 t)e^{-2t} + 2c_2 e^{-2t}, x_2 = c_1 e^{-2t} + c_2 t e^{-2t}$

b. $x_1 = c_1 e^{2t} - c_2 t e^{2t}, x_2 = (c_1 + c_2 t)e^{2t} + 2c_2 e^{-2t}$

c. $x_1 = (c_1 + c_2 t)e^{2t} + 2c_2 e^{2t}, x_2 = (c_1 + c_2 t)e^{2t} + c_2 e^{2t}$

d. $x_1 = (c_1 + c_2 t)e^{-2t} + 2c_2 e^{-2t}, x_2 = (c_1 + c_2 t)e^{-2t} + c_2 e^{-2t}$

e. none of the above

24. Find the general solution of the system:

$x' = 3x + 2y, \qquad y' = -5x + y$

a. $x = 2e^t(c_1 \cos 3t + c_2 \sin 3t)$
$y = e^{2t}\left[c_1(-\cos 3t - 3\sin 3t) + c_2(\cos 3t - \sin 3t)\right]$

b. $x = e^t(c_1 \cos 3t + c_2 \sin 3t)$
$y = e^{2t}\left[c_1(-\cos 3t - 3\sin 3t) + c_2(\cos 3t - \sin 3t)\right]$

c. $x = e^t(c_1 \cos 3t + c_2 \sin 3t)$
$y = 3e^{2t}\left[c_1(-\cos 3t - 3\sin 3t) + c_2(\cos 3t - \sin 3t)\right]$

d. $x = 2e^t(c_1 \cos 3t + c_2 \sin 3t)$
$y = e^{2t}\left[c_1(-\cos 3t - 3\sin 3t) + c_2(3\cos 3t - \sin 3t)\right]$

e. none of the above

25. Find the family of all nonzero solution of equation

$$\frac{d^2y}{dx^2} = \left(\frac{dy}{dx}\right)^2$$

a. $y = \dfrac{1}{x + c_1} + c_2$

b. $y = c_2 + \log(c_1 - x)$

c. $y = c_2 - \log(c_1 + x)$

d. $y = c_2 - \log(c_1 - x^2)$

e. none of the above

26. Solve the initial value problem $y'' + 4y' + 4y = 0,\ y(0) = 1, y'(0) = 1$.

a. $y = (1 - 3x)e^{-2x}$

b. $y = (1 + 3x)e^{-2x}$

c. $y = (1 - 3x)e^{2x}$

d. $y = (1 + 3x)e^{2x}$

e. none of the above

27. Solve for y, $(D^2 - 2D + 1)y = 2xe^x$.

a. $y = c_1e^x + c_2xe^x + x^3e^x$

b. $y = c_1e^x + c_2xe^x + \frac{x^2}{3}e^x$

c. $y = c_1e^x + c_2xe^x + \frac{x^3}{4}e^x$

d. $y = c_1e^x + c_2xe^x + \frac{x^3}{3}e^x$

e. none of the above

28. Determine the Wronskian of the fundamental set $\{e^x \cos 2x, e^x \sin 2x\}$

a. $2e^{2x}$

b. $2e^x(x\cos 2x - \sin 2x)$

c. $e^{2x}(\cos 2x - \sin 2x)$

d. $e^x(x\cos^2 2x + \sin^2 2x)$

e. none of the above

29. Use the method of variation of parameter to determine the general solution of

$$y'' - 2y' + y = \frac{e^x}{x^2 + 1}.$$

a. $y = c_1e^x + c_2xe^x + \frac{1}{2}e^x \ln(1 + x^2) + e^x \tan^{-1} x$

b. $y = c_1e^x + c_2xe^x - \frac{1}{2}e^x \ln(1 + x^2) - xe^x \tan^{-1} x$

c. $y = c_1e^x + c_2xe^x - \frac{1}{2}e^x \ln(1 + x^2) + xe^x \tan^{-1} x$

d. $y = c_1e^x + c_2xe^x - \frac{1}{2}e^x \ln(1 + x^2) + e^x \tan^{-1} x$

e. none of the above

26. Solve the initial value problem $y'' + 4y' + 4y = 0,\ y(0) = 1, y'(0) = 1$.

a. $y = (1 - 3x)e^{-2x}$

b. $y = (1 + 3x)e^{-2x}$

c. $y = (1 - 3x)e^{2x}$

d. $y = (1 + 3x)e^{2x}$

e. none of the above

27. Solve for y: $(D^2 - 2D + 1)y = 2xe^x$.

a. $y = c_1e^x + c_2xe^x + x^3e^x$

b. $y = c_1e^x + c_2xe^x + \frac{x^2}{3}e^x$

c. $y = c_1e^x + c_2xe^x + \frac{x^3}{4}e^x$

d. $y = c_1e^x + c_2xe^x + \frac{x^3}{3}e^x$

e. none of the above

28. Determine the Wronskian of the fundamental set $\{e^x \cos 2x, e^x \sin 2x\}$

a. $2e^{2x}$

b. $2e^x(x\cos 2x - \sin 2x)$

c. $e^{2x}(\cos 2x - \sin 2x)$

d. $e^x(x\cos^2 2x + \sin^2 2x)$

e. none of the above

29. Use the method of variation of parameter to determine the general solution of $y'' - 2y' + y = \frac{e^x}{x^2 + 1}$.

a. $y = c_1e^x + c_2xe^x + \frac{1}{2}e^x \ln(1 + x^2) + e^x \tan^{-1} x$

b. $y = c_1e^x + c_2xe^x - \frac{1}{2}e^x \ln(1 + x^2) - xe^x \tan^{-1} x$

c. $y = c_1e^x + c_2xe^x - \frac{1}{2}e^x \ln(1 + x^2) + xe^x \tan^{-1} x$

d. $y = c_1e^x + c_2xe^x - \frac{1}{2}e^x \ln(1 + x^2) + e^x \tan^{-1} x$

e. none of the above

30. Which of the following given system does the corresponding solution(s) is/are not true?

a. $\mathbf{x}' = \begin{pmatrix} -3 & 2 \\ -3 & 4 \end{pmatrix}\mathbf{x}$; solutions $\mathbf{x}_1 = \begin{pmatrix} e^{3t} \\ 3e^{3t} \end{pmatrix}$ and $\mathbf{x}_2 = \begin{pmatrix} e^{-t} \\ -e^{-t} \end{pmatrix}$

b. $\mathbf{x}' = \begin{pmatrix} 3 & -1 \\ 5 & -3 \end{pmatrix}\mathbf{x}$; solutions $\mathbf{x}_1 = \begin{pmatrix} e^{2t} \\ e^{2t} \end{pmatrix}$ and $\mathbf{x}_2 = \begin{pmatrix} e^{-2t} \\ 5e^{-2t} \end{pmatrix}$

c. $\mathbf{x}' = \begin{pmatrix} 4 & 1 \\ -2 & 1 \end{pmatrix}\mathbf{x}$; solutions $\mathbf{x}_1 = \begin{pmatrix} e^{3t} \\ -e^{3t} \end{pmatrix}$ and $\mathbf{x}_2 = \begin{pmatrix} e^{2t} \\ -2e^{2t} \end{pmatrix}$

d. $\mathbf{x}' = \begin{pmatrix} -3 & 1 \\ -1 & -1 \end{pmatrix}\mathbf{x}$; solutions $\mathbf{x}_1 = \begin{pmatrix} e^{-2t} \\ -e^{-3t} \end{pmatrix}$ and $\mathbf{x}_2 = \begin{pmatrix} e^{-t} \\ e^{-t} \end{pmatrix}$

e. none of the above

31. Find $L^{-1}\left[\dfrac{3s-2}{s^2+4s+20}\right]$

a. $e^{2t}(3\cos(4t) - 2\sin 4t)$

b. $e^{-2t}(3\cos(4t) + 2\sin 4t)$

c. $e^{2t}(3\cos(4t) + 2\sin 4t)$

d. $e^{-2t}(3\cos(4t) - \sin 4t)$

e. none of the above.

32. Determine $L[f]$ if $f(x) = \begin{cases} 0 & \text{for } 0 \le t \le 1 \\ 1+5t & \text{for } t \ge 1 \end{cases}$

a. $\left[\dfrac{5}{s^2} + \dfrac{1}{s}\right]e^{-s}$

b. $\left[\dfrac{5}{s^2} + \dfrac{6}{s}\right]e^{-s}$

c. $\dfrac{5}{s^2} + \dfrac{6}{s}$

d. $\left[\dfrac{5}{s^2} + \dfrac{6}{s}\right]e^{s}$

e. none of the above

33. Find $L^{-1}\left[\dfrac{se^{-3s}}{s^2+4}\right]$

a. $\cos(2t)\,H(t-3)$

b. $\cos(2t-3)\,H(t-3)$

c. $\cos[(t-3)]\,H(t-3)$

d. $\cos[2(t-3)]\,H(t-3)$

e. none of the above.

34. For $(x^2-1)^2 y'' - (x-1)y' + 3y = 0$, $x = 1$ is

a. ordinary point

b. regular singular point

c. irregular singular point

d. irregular point

e. none of the above.

5. For $x^2y''-4xy'-2xy=0$, $x=0$ is

a. ordinary point
b. regular singular point
c. irregular singular point
d. irregular point
e. none of the above.

For question 36 and 37 find the indicial equation given at the indicated singularity.

6. $x^2y''+4xy'+2y=0$ at $x=0$

a. $s^2+3s+2=0$
b. $s^2+3s-2=0$
c. $s^2-s-3=0$
d. $s^2-s+3=0$
e. none of the above.

7. $4x\sin xy''-3y=0$ at $x=0$

a. $s^2+3s+2=0$
b. $s^2+3s-2=0$
c. $s^2-s-3=0$
d. $s^2-s+3=0$
e. none of the above.

Use the statement below to answer questions 38 and 39.

$\mathbf{x}(t)=\begin{pmatrix} a_1t+b_1 \\ a_2t+b_2 \end{pmatrix}$ is a solution of $\mathbf{x}'=\begin{bmatrix} 4 & 2 \\ 3 & -1 \end{bmatrix}\mathbf{x}+\begin{bmatrix} 8t \\ 2t-3 \end{bmatrix}$

8. What are the values of a_1 and a_2?

a. $a_1=\frac{2}{5}$ and $a_2=-\frac{2}{25}$
b. $a_1=\frac{2}{5}$ and $a_2=\frac{2}{25}$
c. $a_1=-\frac{6}{5}$ and $a_2=\frac{8}{5}$
d. $a_1=-\frac{6}{5}$ and $a_2=-\frac{8}{5}$
e. none of the above